Wearable Alert Device: Final Budget

University of Oklahoma
School of Electrical and Computer Engineering
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Senior Capstone Project

<u>Team Members:</u>
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Written By:
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Item	Item Description	Quantity	Cost $	Use	Part Number
1	Arduino Nano ESP32	2	$38.60	Main Micro controller for project	Mouser - ABX00083
2	Lithium Ion Battery Pack - 3.7V 4400mAh	2	$39.90	Main Power supply for project	Adafruit - 354
3	Micro-Lipo Charger for LiPoly Batt with USB Type C Jack	2	$11.90	Rechargable module for Battery Pack	Adafruit - 4410
4	Gravity: GNSS GPS BeiDou Receiver Module - I2C&UART	2	$35.80	GPS Module for Project	Digikey - 1738-TEL0157-ND
5	FERMION MEMS MICROPHONE MODULE(Analog)	2	$5.80	Microphone Module for GS Detection	Digikey - SEN0487
6	320E11RED	2	$4.18	Button for panic device	Digikey - 101391
7	MIC2288YD5-TR	2	$1.46	Booster DC DC Converter	Mouser - MIC2288YD5
8	Breakout Board	2	$2.50	Breakout board for Booster converter	Digikey - 1568-00717-ND
9	SCHOTTKY RECTIFIER	3	$0.87	Schottky diode for GPS Module	Digikey -497-11370-1-ND - Cut Tape (CT)
10	Testing LED (Red)	1	$0.55	LED used to tell user if device is working	Digikey - SLR-56VC3F
11	Decupling Capacitor (0.1 Microfarad)	1	$0.89	Used to remove noise from battery pack power	Digikey - C322C104K3G5TA
12	Current Limiting Resistor(1 KOhm)	1	$0.10	Prevents blow out of LED	Digikey - CFF14JT1K00
13	Wires(Jumper/Cut) (Spool)	1	$15.89	Came with boards but used to connect device	Half came with the Micro breadboards/ other half from link
14	Micro Breadboards	2	$1.98	Used to seat device	6 Pcs Mini Small Breadboard kit White 170 Tie Points
15	Proto Boards	2	$0.56	Used to more perminently seat device	ELEGOO 32 Pcs Double Sided PCB Board Prototype Kit
16	Electrical Tape(Roll)	1	$0.97	Used to hold device together before case	Hyper Tough Electrical Tape Single Roll 60 Feet Long Black
		Total:	$161.95		